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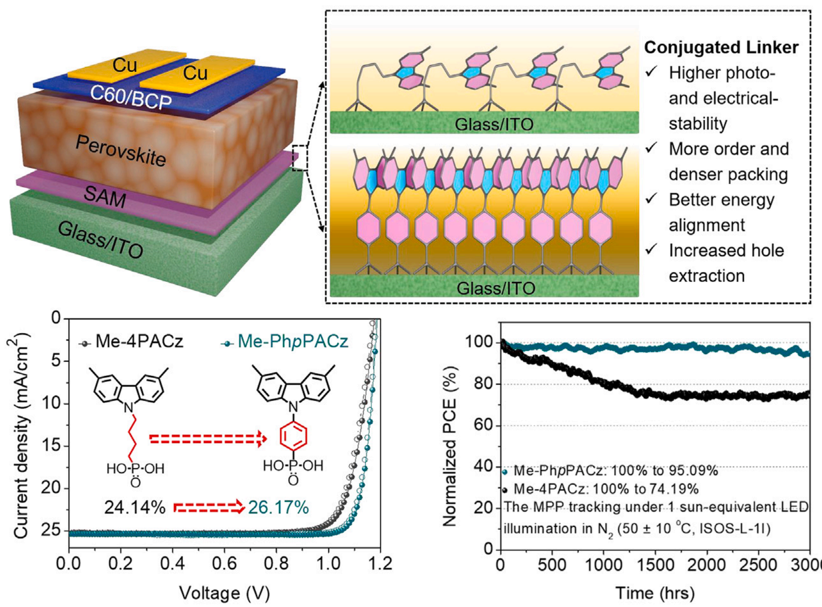

Geping Qu, Siyuan Cai, Ying Qiao, ..., Xihan Chen, Alex K.-Y Jen, Zong-Xiang Xu

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Highlights

Enhancing self-assembled monolayer molecular engineering with phenylene linker

Confirmation of molecular structure's influence on SAM molecules arrangement

Ultrafast spectroscopy reveals phenylene linker greatly enhances SAM hole extraction

Achieving a power conversion efficiency of 26.17% with high stability

This study improves on commercial self-assembled monolayers (SAMs) like Me-4PACz by replacing its linker with a conjugated phenylene, creating Me-Ph*p*PACz for inverted perovskite solar cells (PSCs). The resulting PSCs displayed a record power conversion efficiency (PCE) of 26.17%, along with a fill factor (FF) of 86.79% and exceptional stability. Ultrafast spectroscopy showed that Me-Ph*p*PACz not only regulates the electrode's work function (W_F) but also aids carrier extraction and transport. Our in-depth analysis enhances our understanding of the intricate relationship between SAM molecular design and PSC performance, opening up opportunities for further boosting PSC performance.

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Conjugated linker-boosted self-assembled monolayer molecule for inverted perovskite solar cells

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SUMMARY

This study builds upon the success achieved from commercial self-assembled monolayers (SAMs), such as Me-4PACz, by replacing its flexible butyl linker with a conjugated phenylene ring to form a SAM molecule, Me-Ph*p*PACz, for inverted perovskite solar cells (PSCs). The Me-Ph*p*PACz-derived PSCs exhibited an unprecedented power conversion efficiency of 26.17% with an extremely high fill factor of 86.79% and exceptional long-term stability. The result from ultrafast spectroscopy shows that phenylene-linked SAMs not only modulate the work function of indium tin oxide electrode but also facilitate charge carrier extraction and transport. Our detailed study helps deepen our understanding of the delicate relationship between SAM molecular design and PSC performance, paving the way for further enhancing the efficiency and stability of PSCs.

INTRODUCTION

Recently, remarkable advancements have been made in perovskite solar cells (PSCs), with power conversion efficiencies (PCEs) surpassing 26%.[1] Further improvement of PSCs requires the delicate control of the interfaces between the absorber and charge-transporting layers (CTLs). Since there are only limited choices of inorganic materials to be used as proper interlayers for PSCs,[2,3] exploring synthetic organic chemistry to fine-tune material properties in device engineering has been identified as a key for further enhancing the performance of PSCs. Consequently, the perovskite research community has developed numerous organic semiconductors as CTLs for PSCs, including the very promising self-assembled monolayers (SAMs) to serve as a very effective hole-selective layer (HSL) for inverted (p-i-n) PSCs. From a molecular structure perspective, a typical SAM molecule comprises an anchoring group, a linker, and a terminal group. Each of these elements serves a distinct role in enhancing PSC performance.[4] Phosphonic acid (PA) has been found to exhibit very strong binding energy to enhance the stability and performance of SAMs.[5,6] Terminal groups, such as carbazole (Cz) derivatives, contribute to surface coverage, modify surface properties, and generate a dipole moment that can tune the substrate's work function (W_F).[4,7] A widely studied SAM molecule, [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]PA (Me-4PACz), has been extensively used to achieve excellent performance in inverted PSCs.[8,9] However, there is still a lot of room for further improving molecular design to push PCE closer to the Shockley-Queisser (S-Q) limit. The linker group, which connects the terminal group and anchoring group, has a significant influence on the self-assembly process. Although flexible alkyl chains are commonly used in SAMs for PSCs, they are not the optimal

CONTEXT & SCALE

Self-assembled monolayer (SAM) has become an indispensable component in perovskite solar cells (PSCs) to enhance performance and reduce cost. However, concerns persist about the efficiency and stability of SAM molecules. Additionally, the impact of SAM molecular structure on thin film packing and hole extraction or transport ability on PSC performance remains obscure. Here, we developed a novel SAM, Me-Ph*p*PACz, by replacing the flexible alkyl linker with a conjugated phenylene linker in the commercial SAM Me-4PACz. This modification enhances the molecule's aspect ratio and dipole moment, resulting in improved film coverage and hole extraction. Consequently, the power conversion efficiency and stability of inverted PSCs have significantly improved, reaching 26.17% for Me-Ph*p*PACz compared with 24.14% for Me-4PACz. Moreover, conjugated linkers can also be applied to various carbazole-based SAMs, yielding higher performance compared with non-conjugated alkyl chain-connected SAM molecules.

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choice due to their insulating property and flexibility.[10] These factors can hinder charge flow and intermolecular interactions, in addition to affecting the molecular density on the substrate. Therefore, it is rational to employ a conjugated, rigid linker to provide a more robust and conductive backbone to offer improved stability and rapid charge transfer capabilities.[10–12]

Because SAMs are ultrathin, their compactness and coverage are crucial for efficient charge extraction and minimized shunt paths.[13] Although the co-adsorption of two different SAM molecules has been suggested to fill possible voids,[13,14] the optimal approach is to modify the best available SAMs, such as Me-4PACz. After a combined theoretical and experimental analysis, we developed a SAM, [4-(3,6-dimethyl-9H-carbazol-9-yl)phenyl]PA (Me-Ph*p*PACz), by replacing the flexible alkyl linker with a rigid, planar, and conjugated phenylene linker. This modification enhances the molecule's aspect ratio, dipole moment, and charge transfers, which improves the film coverage and hole transport. As a result, the PCE and stability of inverted PSCs have significantly improved, reaching 26.17% for Me-Ph*p*PACz compared with 24.14% for Me-4PACz.

RESULTS AND DISCUSSION

The detailed synthesis of SAM molecules is depicted in the supplemental information (Figure S1), and all the synthesized key intermediates and the final compounds were characterized by ^{1}H nuclear magnetic resonance (NMR), ^{13}C NMR, ^{31}P NMR, and high-resolution mass spectrometry to examine their molecular structures and purity (Figures S2–S37). Cyclic voltammetry (CV) and UV-vis spectra were also performed to study their electrochemical and optical properties (Figures S38 and S39; Table S1). All SAM materials exhibited p-type semiconductor properties, which are favorable for application in HSL of inverted PSC.

By replacing an alkyl linker with a phenylene linker, it helps improve the molecular rigidity, materials stability, packing density of the SAMs, and their capability in hole extraction. Calculation analysis of Me-4PACz (Figure 1A) revealed that the highest occupied molecular orbitals (HOMOs) and lowest unoccupied molecular orbitals (LUMOs) of Me-4PACz are both located on the electron-rich Cz terminal group, and the LUMO is as high as −0.95 eV, which results in high photoactivity. Although the HOMO and LUMO orbitals of Me-Ph*p*PACz (Figure 1B) are more effectively delocalized and spatially separated, which helps to enhance the photo- and electrical stability induced by effective electron/charge delocalization and stabilized LUMO energy level (−1.4 eV).[12] These effects are reflected in the CV measurements. The current densities of Me-4PACz decreased significantly during 25 cycles of CV, suggesting lower electrochemical stability compared with Me-Ph*p*PACz (Figures 1C and 1D). For measuring photostability, SAM molecules were dissolved in deuterated dimethyl sulfoxide (DMSO-d_6) and exposed to 1 sun equivalent illumination for 12 h. The ^{1}H NMR spectra showed significant decomposition in Me-4PACz solutions, with peaks emerging at low field (7.3–8.7 ppm) indicating cleavage of Cz fragments (Figure 1E). On the contrary, minimal changes in color and ^{1}H NMR signals were observed for Me-Ph*p*PACz solutions, confirming its higher photostability (Figures 1F and 1G). The results from thermogravimetric analysis (Figure S40) showed that the conjugated linker had minimal impact on the thermal stability of SAMs, both revealing decomposition temperatures exceeding 300°C.

The interaction between the anchoring group of SAM molecules and indium tin oxide (ITO) was analyzed using X-ray photoelectron (XPS) spectroscopy (Figure S41;

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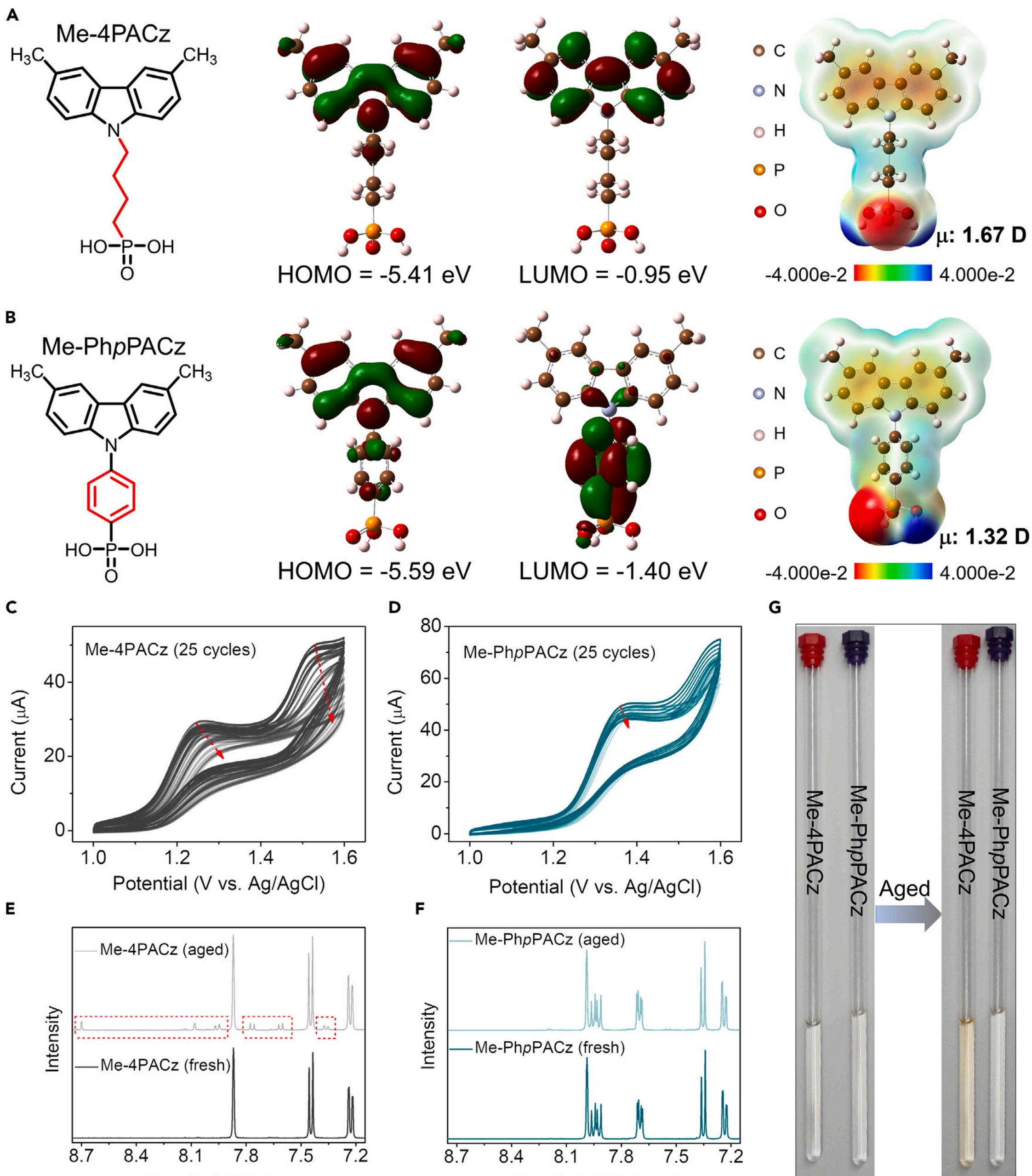

Figure 1. Conjugated structures for improving the stability of SAM molecules

(A and B) Chemical structures, frontier molecular orbitals with energy level diagram, electrostatic potential surfaces (EPSs), and dipole moment (μ) of (A) Me-4PACz and (B) Me-Ph*p*PACz.

(C and D) Repeated CV trace (25 cycles) of (C) Me-4PACz and (D) Me-Ph*p*PACz.

(E and F) ^{1}H NMR spectra evolution of (E) Me-4PACz and (F) Me-Ph*p*PACz.

(G) Photographs of SAM materials solution before and after light aging.

Table S2). The O 1s spectra of Me-4PACz and Me-Ph*p*PACz powders showed two peaks at 533 and 532 eV, corresponding to P–O–H and P = O,[15,16] respectively (Figures 2A and 2B). The ITO O 1s spectrum had three fitting peaks at 531.77, 530.03, and 528.84 eV, representing surface oxygen vacancies, In–O, and Sn–O, respectively.[10,17] The O 1s spectra of SAM films on ITO displayed multiple fitting

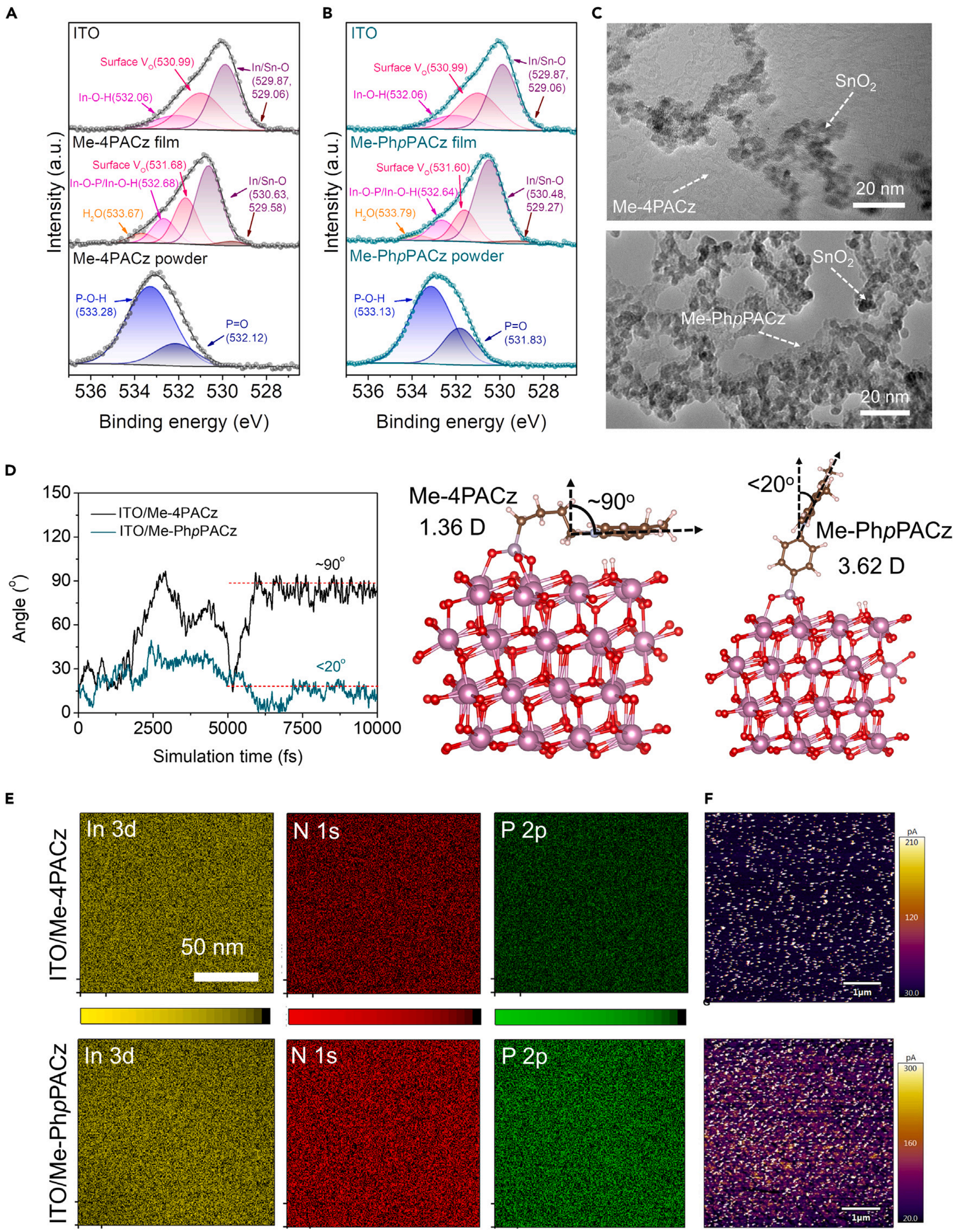

Figure 2. Interaction between SAMs and ITO

(A and B) O 1s XPS spectra are presented for bare ITO, powder, and films of (A) Me-4PACz and (B) Me-Ph*p*PACz.

(C) TEM images of SAMs with SnO_2 particles.

(D) Angular evolution of SAM molecules on ITO surface during AIMD simulations at 300 K, along with the final configurations of SAM molecules on ITO surface.

(E) XPS mapping images of SAM molecules on ITO (In represents ITO, and N and P represent SAMs).

(F and G) Conductive AFM images of (F) Me-4PACz and (G) Me-Ph*p*PACz on ITO.

peaks. In the films, the peaks for P–O–H and P = O in the powder spectra disappeared, and new peaks emerged at 532.68 eV (Me-4PACz) and 532.64 eV (Me-Ph*p*PACz), indicating P–O–In bond formation with ITO.[18] Furthermore, the P 2p spectra showed a decreased phosphorus binding energy in the SAM films (Figure S42). These findings are consistent with the Fourier-transform infrared spectroscopy (FTIR) results (Figure S43 and Note S1), confirming the tridentate anchoring of the PA groups on ITO. Static density functional theory (DFT) calculations (Figures S44 and S45 and Note S2) reveal high binding energies of around −1.80 eV for both SAM molecules on ITO, indicating strong chemical bonding between the PA groups and the ITO surface. However, the results from transmission electron microscopy (TEM) (Figures 2C, S45D, and S45E) emphasize the role of the linker in intermolecular interactions and backbone strength. Although Me-4PACz has high binding energy to SnO_2, some Me-4PACz molecules do not anchor on SnO_2. By contrast, Me-Ph*p*PACz assembles densely on SnO_2 particles, with no free Me-Ph*p*PACz observed in vacant positions. This can be attributed to its rigid and conjugated backbone, reducing molecular twisting, and enhancing intermolecular interactions, which are not observed in Me-4PACz-based SAM.

To explore the role of the rigid ring in Me-Ph*p*PACz, *ab initio* molecular dynamics (AIMD) simulations were employed to analyze the dynamic behavior of SAM molecules on the ITO surface (Figure S46 and Note S3). Compared with Me-Ph*p*PACz, the flexible linker in Me-4PACz led to twisting and bending toward the ITO substrate. Figure 2D illustrates the angular evolution of SAM molecules on the ITO surface during a 10 ps AIMD simulation at 300 K. The angle between the Cz group and the z axis was defined as the orientation of SAM molecules relative to the ITO surface. Due to the presence of the phenylene, Me-Ph*p*PACz maintained a rigid assembly with an angle of $<20^\circ$ relative to the ITO surface. By contrast, Me-4PACz exhibited a twisted assembly on the ITO surface, with the Cz group nearly parallel to the ITO substrate. To experimentally verify this bending and twisting, X-ray reflectivity (XRR) measurements were performed on Me-4PACz and Me-Ph*p*PACz thin films, revealing quite different SAM layer heights of 3.99 and 10.00 Å, respectively. These values were consistent with the theoretically calculated molecular lengths obtained after AIMD simulations: 4.00 ± 0.14 Å for Me-4PACz and 10.58 ± 0.23 Å for Me-Ph*p*PACz (Figure S47 and Note S4). After bending and twisting, molecules will have weaker dipole moments, affecting the alignments and uniformity of SAM. However, the rigid structure of Me-Ph*p*PACz promotes uniform and denser molecular packing and enhanced dipole moments from 1.32 to 3.62 D. This is supported by the observations in Figures 2A and 2B and Table S3, where the Me-4PACz film showed more oxygen vacancies, indicating less interaction with ITO. Additionally, XPS mapping revealed weaker N and P intensities on the ITO surface for Me-4PACz than those for Me-Ph*p*PACz (Figure 2E). These observations suggest that Me-Ph*p*PACz's vertical alignment orientation leads to increased packing density on ITO. This is further confirmed by the quantitative analysis of SAMs on ITO through XPS full spectra and fine spectra of nitrogen, phosphorus, and indium, as well as the larger water contact angle of Me-Ph*p*PACz compared with Me-4PACz (Figures S41, S48, and S49; Tables S2 and S4; and Notes S5 and S6). Furthermore, the influence of linking position is investigated by preparing the anchoring group at the *m*-position of the linking ring, Me-Ph*m*PACz (Figures S50 and S51 and Note S3). This molecule also displays a lying-down molecular configuration. It is well established that denser molecular packing usually leads to smoother surfaces, and aligned packing will contribute to smaller potential differences. These observations are corroborated by our atomic force microscopy (AFM) measurements, with root mean square (RMS) roughness values for Me-4PACz and Me-Ph*p*PACz on ITO amounting to 4.0

and 3.7 nm, respectively (Figure S52 and Note S7). The vertically arranged conjugated SAMs create a highly dense and ordered HSL substrate, which greatly facilitates the perovskite growth.[19,20] This is validated by the improved perovskite quality grown on Me-Ph*p*PACz compared with Me-4PACz, evident by photo images, photoluminescence (PL) mapping, scanning electron microscope (SEM) images, and X-ray diffraction spectroscopy (Figures S53–S57 and Notes S8–S10).

To explore the role of conjugated linkers on perovskite interactions, adsorption energies were calculated using DFT calculations (Figure S54 and Note S8). The results reveal that perovskite exhibits a higher adsorption affinity for Me-Ph*p*PACz (−1.48 eV) compared with Me-4PACz (−0.23 eV). Improvement in carrier transport is also corroborated by conductive AFM (*c*-AFM). The maps of both SAMs on ITO are presented in Figures 2F and 2G, visualized through brightness contrast to assess their electrical performance. Compared with Me-4PACz, Me-Ph*p*PACz exhibits higher brightness contrast at larger current values, with an average current of approximately 120 and 160 pA, indicating better charge-transport characteristics within the film. Me-Ph*p*PACz leads to clearer ITO SEM images, suggesting enhanced HSL conductivity (Figure S53). As shown in Figure 3A, the increased dipole moment of Me-Ph*p*PACz SAM must have a positive impact on their energy alignment and, thus, on hole extraction and transport. The spatial charge-limited current measurements revealed a slightly reduced trap state density (3.22×10^{15} cm^{-3}) and an increased hole mobility (1.15×10^{-3} $cm^2V^{-1}s^{-1}$) for Me-Ph*p*PACz sample compared with Me-4PACz ($n_{trap} = 3.68 \times 10^{15}$ cm^{-3} and $\mu_h = 8.4 \times 10^{-4}$ $cm^2V^{-1}s^{-1}$) (Figure S58 and Note S11).

The influence of dipole moment on energy levels, investigated using UV-vis spectroscopy and ultraviolet photoelectron spectroscopy (UPS) (Figures S59 and S60; Table S5 and Note S12), is schematically depicted in Figure 3B. The orientation of the dipole moment is crucial for achieving a beneficial modification of ITO's W_F, which can be effectively realized if the molecules are suitably organized. This is exemplified by Me-Ph*p*PACz, which restricts bending and maintains a more perpendicular configuration, supported by a rigid linker. The W_F of ITO/Me-Ph*p*PACz (−4.65 eV) is notably deeper than the values measured for the bare ITO (−4.20 eV) and ITO/Me-4PACz (−4.55 eV), enabling it to be more efficient for hole extraction and further generating a potential barrier at the perovskite conduction band interface for electron injection. The SAMs also influence the energy level of perovskite. The W_F of perovskite changes to be closer to the valence band, from −4.12 to −4.23 and −4.64 eV when deposited on ITO, Me-4PACz, and Me-Ph*p*PACz, respectively, which can be more favorable for hole transport. Additionally, the energy offsets between the perovskite valence band edge and the W_F are 1.53, 1.41, and 0.91 eV for ITO/PVK, ITO/Me-4PACz/PVK, and ITO/Me-Ph*p*PACz/PVK, respectively. This suggests that the energy loss for hole extraction is minimal at the Me-Ph*p*PACz/perovskite interface. The larger energy offset between the perovskite conduction band edge and the W_F of ITO/Me-Ph*p*PACz/PVK in comparison to ITO/PVK and ITO/Me-4PACz/PVK could provide better blocking of the electron injection from the perovskite, leading to reduced carrier recombination and enhanced fill factor (FF).[21,22]

The introduction of a conjugated linker enhances the charge distribution, as shown by the charge density distributions between Me-Ph*p*PACz and the perovskite, which span the entire interface. By contrast, Me-4PACz and the perovskite only exhibit prominent localized features (Figure S61). Furthermore, we noted an irregularly discontinuous charge distribution in the lower part of the alkyl linker region in the

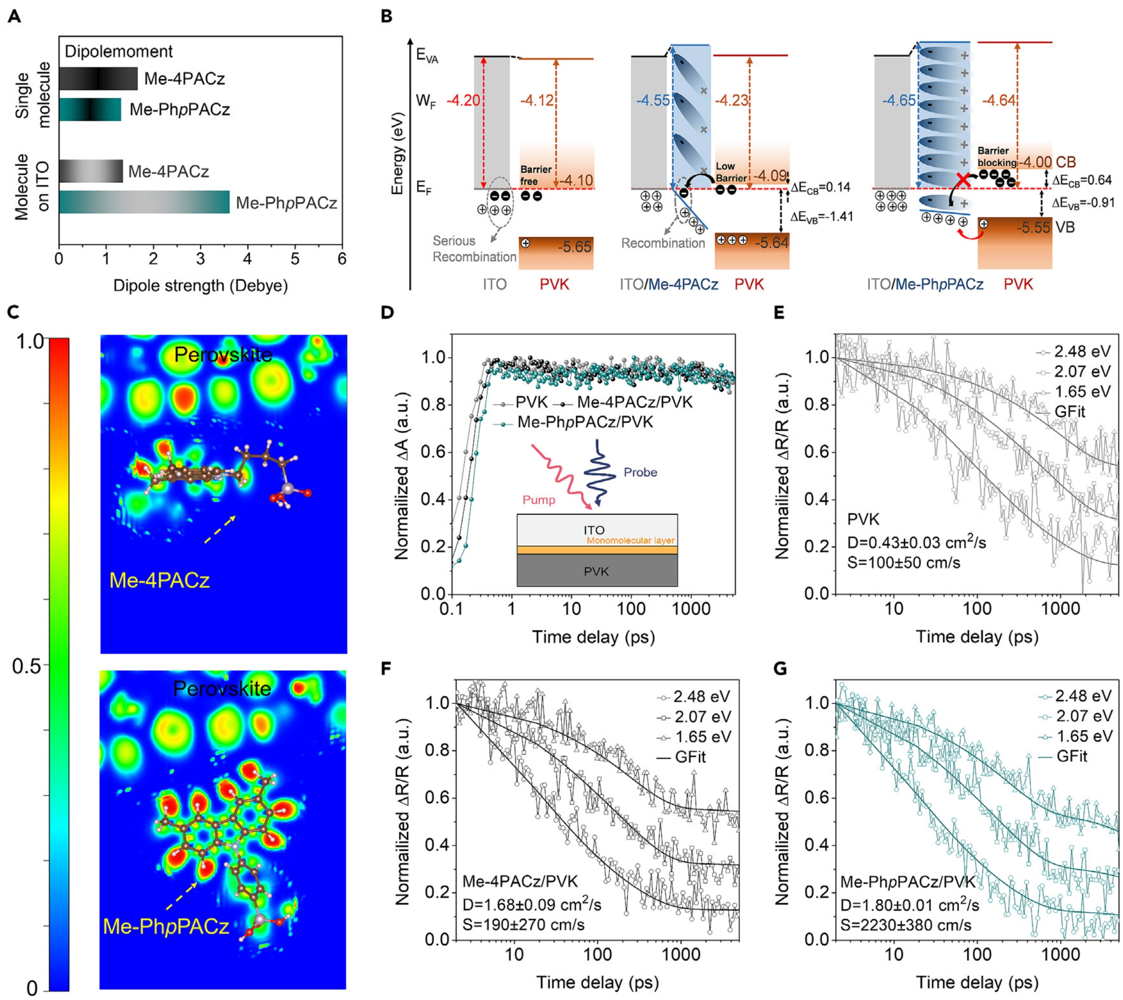

Figure 3. Different linker groups in SAMs exhibit varied hole transport mechanisms

(A) Molecular dipole moments of Me-4PACz and Me-Ph*p*PACz, along with their dipole moments on ITO.

(B) Schematic representation illustrating the influence of energy levels and dipole moments of ITO and perovskite films on charge carrier transport (E_{VA} represents the vacuum energy level, E_F is the Fermi level, W_F is the work function, CB is the conduction band, and VB is the valence band).

(C) ELF in the SAM molecular region on the perovskite surface.

(D) Schematic illustration of ultrafast pump-probe transient absorption (TA) kinetics measurements and TA kinetics of perovskites excited with 1.65 eV at 10 × 17 cm^{-3} carrier concentration.

(E–G) Surface carrier kinetics from excitation at 2.48, 2.07, and 1.65 eV probed by transient reflection (TR) spectroscopy and fitted with a diffusion-surface extraction model for (E) perovskite, (F) Me-4PACz/perovskite, and (G) Me-Ph*p*PACz/perovskite.

Me-4PACz molecule, as evident from the electron localization function (ELF) diagram (Figure 3C). By contrast, the Me-Ph*p*PACz molecule with a phenylene linker exhibits continuous charge distributions between the molecule and the perovskite surface. This facilitates photo-induced charge carriers more effective transport along the in-plane direction to the ITO.

Ultrafast spectroscopic techniques are effective in probing the transport and recombination properties of charge carriers in PSCs.[23,24] As illustrated in Figure 3D, the concentration of bulk carriers shows negligible decay over a time scale of 5,000 ps, suggesting that the bulk carrier recombination process is insignificant. By globally fitting the carrier dynamics obtained from pump pulses of three wavelengths, we can derive carriers' diffusion coefficients (D) and surface extraction velocities (S) (Figures 3E–3G

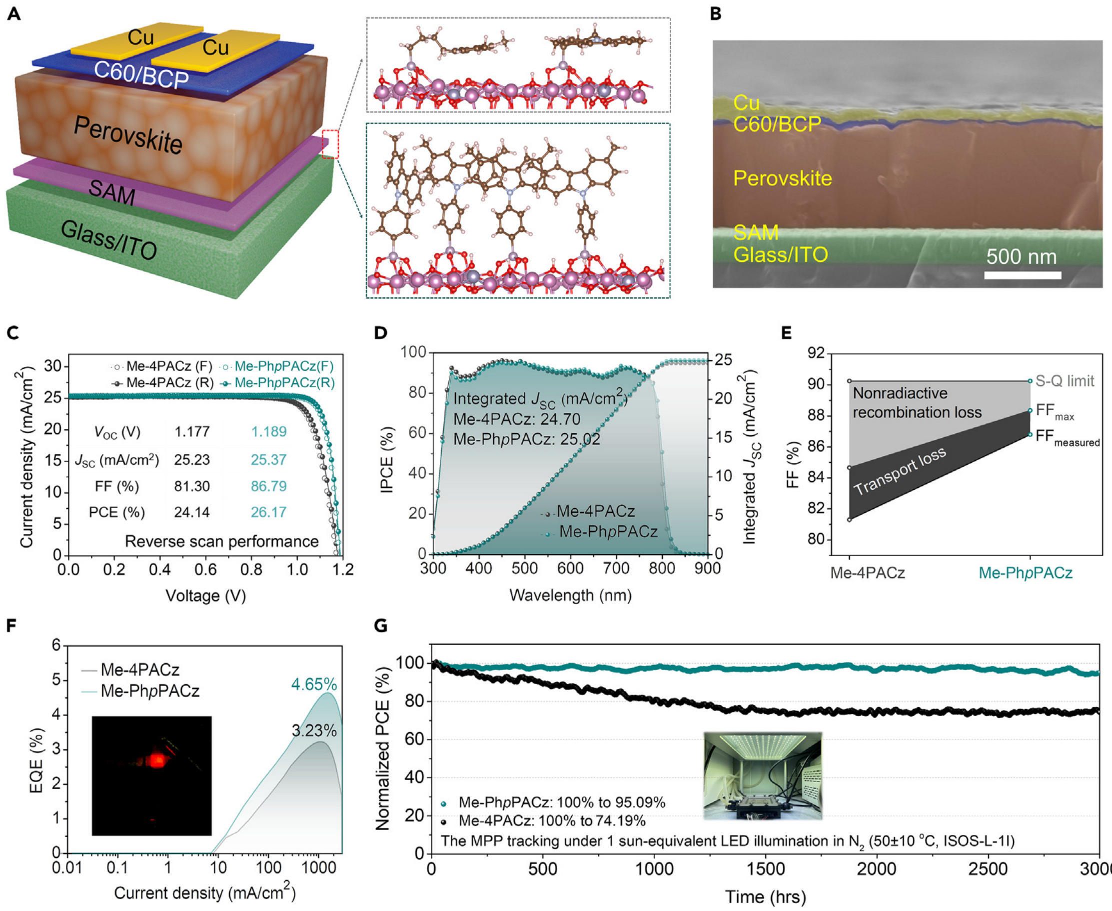

Figure 4. Photovoltaic performance of inverted perovskite solar cells

(A) Device structure and the final configurations of multiple Me-4PACz and Me-Ph*p*PACz molecules stacking on the ITO surface after a 10 ps AIMD simulation.

(B) Cross-sectional SEM image of the Me-Ph*p*PACz device.

(C–G) Devices performance for (C) J-V curves of Me-4PACz and Me-Ph*p*PACz-based PSCs (Data S1. JV data for the champion device based on Me-Ph*p*PACz), (D) IPCE curves of Me-4PACz and Me-Ph*p*PACz-based PSCs and integrated J_{SC} over the AM 1.5G standard spectrum, (E) The device FF S-Q limit, consisting of charge-transport loss and non-radiative loss, (F) EQEEL of Me-4PACz and MePh*p*PACz-based PSCs in the range of 0.01–2,000 mA/cm^2 and EL images, and (G) MPP tracking under N_2 and simulated 1 sun AM 1.5G illumination for uncooled devices (reaching an operating temperature of 50°C ± 10°C).

and Note S13). The obtained data indicate that the diffusion coefficient of Me-4PACz is nearly four times higher than that of pure perovskite. Interestingly, the surface extraction velocities of Me-4PACz-based perovskite (190 ± 270 cm/s) and bare perovskite (100 ± 50 cm/s) are almost identical, suggesting that the presence of alkyl chains does not significantly improve surface extraction. By contrast, Me-Ph*p*PACz exhibits significantly enhanced surface extraction velocity up to 2,230 ± 380 cm/s compared with the control group, indicating a crucial role of the conjugated structure in the hole extraction process. These results are also consistent with steady PL and time-resolved PL (TRPL) measurements (Figure S62; Table S6; Note S14).

The device structure and cross-sectional SEM images of the inverted PSCs are presented in Figures 4A, 4B, and S63. AIMD simulations reveal the microscopic stacking arrangement of multiple SAM molecules on the ITO surface. Tauc's plot and the first-order derivative of external quantum efficiency (EQE) suggest a perovskite band gap of ~1.55 eV (Figure S64). The *J–V* curves and corresponding photovoltaic parameters for Me-4PACz- and Me-Ph*p*PACz-based champion PSCs are shown in Figure 4C and Table S7. The FF improves from 81.30% for Me-4PACz to 86.79% for Me-Ph*p*PACz. Me-4PACz achieves a champion PCE of 24.14%, while Me-Ph*p*PACz reaches 26.17% (as shown in Figure S65, the certified PCEs were 25.82% with a steady-state efficiency of 25.18% from the Shanghai Institute of Microsystem and Information Technology (SIMIT) and 26.12% with a steady-state efficiency of 24.81% from the National PV Industry Measurement and Testing Center (NPVM), respectively.) for the reverse scan. This enhancement is consistent with the photovoltaic average parameters of 30 devices (Figure S66). The EQE results indicate that the integrated currents of Me-4PACz and Me-Ph*p*PACz are similar, measuring 24.70 and 25.02 mA/cm^2, respectively (Figure 4D). Under AM 1.5G simulated solar irradiation in an N_2 glovebox, both Me-4PACz and Me-Ph*p*PACz devices exhibit excellent stable continuous output performance after 60 min of steady-state testing. The PCE values decrease from the initial values of 24.06% and 26.05% to 23.75% and 25.87%, respectively (Figure S67).

The Me-Ph*p*PACz-based devices demonstrated enhanced performance characteristics, induced by a reduced ideality factor (n_{ID}), increased carrier migration, and minimized resistance, as determined by various characterization methods such as light intensity-dependent measurements of short circuit current density (J_{SC}) and open circuit voltage (V_{OC}), electrochemical impedance spectroscopy (EIS), Mott-Schottky plot analysis, transient photocurrent (TPC), and photovoltage (TPV) (Figures S68–S71 and Notes S15–S18). The two main reasons for FF values below the S-Q limit are trap-assisted non-radiative losses and charge-transport losses. According to n_{ID} calculations, the maximum FF (FF_{max}) for Me-4PACz and Me-Ph*p*PACz is found to be 84.66% and 88.15%, respectively (Figure 4E).[25,26] This finding corroborates the notion that using Me-Ph*p*PACz as HSL at the bottom interface can effectively suppress trap-assisted non-radiative recombination and improve charge-transport within the device.[27,28]

The PSC based on Me-Ph*p*PACz demonstrates superior performance in the external quantum efficiency for electroluminescence (EQEEL), exhibiting higher electroluminescence (EL) emission intensity, which indicates a higher level of carrier injection (Figure S72 and Note S19).[29] The EQEEL values for Me-4PACz-based and Me-Ph*p*PACz-based PSCs are 3.23% and 4.65%, respectively (Figure 4F). Analogous to the EQEEL results, employing PL quantum yield (PLQY) in semi-stacked and full-stacked devices to estimate quasi-Fermi level splitting (QFLS) reveals higher quantum yield and increased QFLS for Me-Ph*p*PACz (Figure S72D), suggesting that optimized bottom interface energy level alignment not only reduces perovskite defects but also minimizes interface non-radiative recombination losses. In accordance with the ISOS-L-1I standard protocol,[30] the performance of unencapsulated PSCs was tracked at the maximum power point (MPP) under continuous illumination in a cyclic nitrogen environment at 50°C $\pm$ 10°C. After more than 3,100 h of continuous exposure, the PSC based on Me-Ph*p*PACz retained 98.70% of its initial PCE, whereas the PSC based on Me-4PACz dropped to 74.07% during the same period (Figure 4G). Additionally, further thermal stability assessments of the encapsulated devices, aged at 85°C in ambient air (RH 75% $\pm$ 15%) and subjected to MPP tracking under continuous 1 sun equivalent LED illumination in ambient air (RH 70% $\pm$ 20%) at an operating temperature of 65°C $\pm$ 10°C, also validate the enhanced stability of Me-Ph*p*PACz (Figure S73).

In terms of the versatility of SAMs with conjugated linkers, the choice of NiO_X as the substrate for Me-Ph*p*PACz in the ITO/NiO_X/SAM/PVK/C_{60}/BCP/Cu-based PSCs structure led to a higher PCE of 25.72%, surpassing Me-4PACz (23.98%) (Figure S74A). Altering the position of the PA anchoring group on the phenylene linker resulted in a PCE of 25.51% for the *m*-substituted Me-Ph*m*PACz (Figure S74B), which is also higher than Me-4PACz with non-conjugated linker. Examining the influence of the Cz terminal group in SAMs, Ph*p*PACz was synthesized and compared with 4PACz, leading to an efficiency increase from 23.68% to 24.85% (Figure S74C). Furthermore, MeO-Ph*p*PACz demonstrated an increase in PCE from 22.78% to 25.21%, outperforming MeO-4PACz (Figure S74D). Expanding the research to different perovskite compositions, the conjugated linker-based SAM of MeO-Ph*p*PACz was incorporated into wide-band-gap perovskites with a composition of $Cs_{0.2}FA_{0.8}PbI_{1.8}Br_{1.2}$ (Figure S75), achieving a PCE of 20.08% with a higher V_{OC} of 1.302 V, which is superior to that of MeO-4PACz (PCE of 18.67% and V_{OC} of 1.268 V). These findings emphasize the adaptability of conjugated linker optimization for SAMs in PSCs.

Conclusions

In this study, we designed and synthesized a SAM material, Me-Ph*p*PACz, which is based on Cz terminal and PA anchoring groups connected by a conjugated phenylene linker. This material was then applied to the HSL of inverted PSCs. The experimental and theoretical simulation results showed that compared with Me-4PACz, which uses butyl chains as a linker, Me-Ph*p*PACz molecules exhibit higher photoelectric stability and are less susceptible to bending and folding during molecular assembly. This facilitates the formation of a high-density, well-ordered SAM film. As a result, the W_F of ITO electrodes is better regulated, and the quality of the perovskite thin film is improved. Ultrafast spectroscopic experiments revealed that Me-Ph*p*PACz-based devices exhibit a more compatible energy level alignment, enhanced device hole extraction, and transport efficiency, effectively reduce interface carrier non-radiative recombination, and achieve a FF_{max} of up to 86.79%. The average efficiency of the Me-Ph*p*PACz PSCs device reached 25.82% $\pm$ 0.35%, surpassing the 23.69% $\pm$ 0.45% of the Me-4PACz device, and the device stability was significantly improved. Moreover, this generalizable molecular design strategy using a conjugated linker can also be applied to different Cz-based SAM molecules, yielding higher device performance compared with non-conjugated alkyl chain-connected SAM molecules.

EXPERIMENTAL PROCEDURES

Resource availability

Lead contact

Further information should be directed to the lead contact, Zong-Xiang Xu (xu.zx@sustech.edu.cn).

Materials availability

All the experimental procedures of the material synthesis are provided in the supplemental information.

Data and code availability

- All data needed to evaluate the conclusions in the paper are present in the paper or the supplemental information.
- All the data are available from the corresponding authors upon reasonable request.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.joule.2024.05.005.

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AUTHOR CONTRIBUTIONS

Z.-X.X., G.Q., and S.C. conceived and designed the research. G.Q. and D.W. carried out the fabrication and major characterization of PSCs. S.C. synthesized and characterized the SAM molecules. L.Z. contributed to the material synthesis. Y.Q. and Q.C. conducted theoretical simulations. D.W. optimized perovskite components and device structure. G.Q. carried out the fabrication and major characterization of WBG PSCs. K.J. conducted the encapsulation of the devices. S.G. and X.C. performed ultrafast spectroscopic measurements. Y.-G.W. performed PL mapping and TEM measurements. D.K., X.C., and A.K.-Y.J. contributed to the analysis and provided advice. G.Q., Y.Q., S.C., and Z.-X.X. wrote the initial draft, and all authors contributed to the final paper.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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REFERENCES

1. National Renewable Energy Laboratory. Best Research-Cell Efficiency Chart. https://www.nrel.gov/pv/cell-efficiency.html.

2. Luo, C., Zheng, G., Gao, F., Wang, X., Zhan, C., Gao, X., and Zhao, Q. (2023). Engineering the buried interface in perovskite solar cells via lattice-matched electron transport layer. Nat. Photon. *17*, 856–864. https://doi.org/10.1038/s41566-023-01247-4.

3. Schulz, P., Cahen, D., and Kahn, A. (2019). Halide Perovskites: Is It All about the Interfaces? Chem. Rev. *119*, 3349–3417. https://doi.org/10.1021/acs.chemrev.8b00558.

4. Li, M., Xie, Y., Lin, F.R., Li, Z., Yang, S., and Jen, A.K.-Y. (2023). Self-assembled monolayers as emerging hole-selective layers enable high-performance thin-film solar cells. Innovation (Camb) *4*, 100369. https://doi.org/10.1016/j.xinn.2022.100369.

5. Ambrosio, F., Martsinovich, N., and Troisi, A. (2012). What Is the Best Anchoring Group for a Dye in a Dye-Sensitized Solar Cell? J. Phys. Chem. Lett. *3*, 1531–1535. https://doi.org/10.1021/jz300520p.

6. Guo, R., Zhang, X., Zheng, X., Li, L., Li, M., Zhao, Y., Zhang, S., Luo, L., You, S., Li, W., et al. (2023). Tailoring Multifunctional Self-Assembled Hole Transporting Molecules for Highly Efficient and Stable Inverted Perovskite Solar Cells. Adv. Funct. Materials *33*, 2211955. https://doi.org/10.1002/adfm.202211955.

7. Ali, F., Roldán-Carmona, C., Sohail, M., and Nazeeruddin, M.K. (2020). Applications of Self-Assembled Monolayers for Perovskite Solar Cells Interface Engineering to Address Efficiency and Stability. Adv. Energy Mater. *10*, 2002989. https://doi.org/10.1002/aenm.202002989.

8. Zheng, X., Li, Z., Zhang, Y., Chen, M., Liu, T., Xiao, C., Gao, D., Patel, J.B., Kuciauskas, D., Magomedov, A., et al. (2023). Co-deposition of hole-selective contact and absorber for improving the processability of perovskite solar cells. Nat. Energy *8*, 462–472. https://doi.org/10.1038/s41560-023-01227-6.

9. Ren, Z., Cui, Z., Shi, X., Wang, L., Dou, Y., Wang, F., Lin, H., Yan, H., and Chen, S. (2023). Poly(carbazole phosphonic acid) as a versatile hole-transporting material for p-i-n perovskite solar cells and modules. Joule *7*, 2894–2904. https://doi.org/10.1016/j.joule.2023.10.014.

10. Timpel, M., Nardi, M.V., Krause, S., Ligorio, G., Christodoulou, C., Pasquali, L., Giglia, A., Frisch, J., Wegner, B., Moras, P., and Koch, N. (2014). Surface Modification of ZnO(0001)–Zn with Phosphonate-Based Self-Assembled Monolayers: Binding Modes, Orientation, and Work Function. Chem. Mater. *26*, 5042–5050. https://doi.org/10.1021/cm502171m.

11. Li, C., Zhang, Z., Zhang, H., Yan, W., Li, Y., Liang, L., Yu, W., Yu, X., Wang, Y., Yang, Y., et al. (2024). Fully aromatic self-assembled hole-selective layer toward efficient inverted wide-bandgap perovskite solar cells with ultraviolet resistance. Angew. Chem. Int. Ed. Engl. *63*, e202315281. https://doi.org/10.1002/anie.202315281.

12. Zhang, S., Wu, R., Mu, C., Wang, Y., Han, L., Wu, Y., and Zhu, W.-H. (2022). Conjugated Self-Assembled Monolayer as Stable Hole-Selective Contact for Inverted Perovskite Solar Cells. ACS Materials. Lett. *4*, 1976–1983. https://doi.org/10.1021/acsmaterialslett.2c00799.

13. Kapil, G., Bessho, T., Sanehira, Y., Sahamir, S.R., Chen, M., Baranwal, A.K., Liu, D., Sono, Y., Hirotani, D., Nomura, D., et al. (2022). Tin–Lead Perovskite Solar Cells Fabricated on Hole Selective Monolayers. ACS Energy Lett. *7*, 966–974. https://doi.org/10.1021/acsenergylett.1c02718.

14. Li, E., Bi, E., Wu, Y., Zhang, W., Li, L., Chen, H., Han, L., Tian, H., and Zhu, W.-H. (2020). Synergistic Coassembly of Highly Wettable and Uniform Hole-Extraction Monolayers for Scaling-up Perovskite Solar Cells. Adv. Funct. Materials *30*, 1909509. https://doi.org/10.1002/adfm.201909509.

15. Qi, H., Zhang, H., Wu, X., Tang, Y., Qian, M., Wang, K., and Qi, H. (2020). Matrix-Free and Highly Efficient Room-Temperature Phosphorescence Carbon Dots towards Information Encryption and Decryption. Chem. Asian J. *15*, 1281–1284. https://doi.org/10.1002/asia.202000063.

16. Thermo Fisher Scientific. Oxygen | XPS Periodic Table – CN. https://www.thermofisher.cn/cn/zh/home/materials-science/learning-center/periodic-table/non-metal/oxygen.html.

17. Paramonov, P.B., Paniagua, S.A., Hotchkiss, P.J., Jones, S.C., Armstrong, N.R., Marder, S.R., and Brédas, J.-L. (2008). Theoretical Characterization of the Indium Tin Oxide Surface and of Its Binding Sites for Adsorption of Phosphonic Acid Monolayers. Chem. Mater. *20*, 5131–5133. https://doi.org/10.1021/cm8014622.

18. Puziy, A.M., Poddubnaya, O.I., Socha, R.P., Gurgul, J., and Wisniewski, M. (2008). XPS and NMR studies of phosphoric acid activated carbons. Carbon *46*, 2113–2123. https://doi.org/10.1016/j.carbon.2008.09.010.

19. Li, Z., Sun, X., Zheng, X., Li, B., Gao, D., Zhang, S., Wu, X., Li, S., Gong, J., Luther, J.M., et al. (2023). Stabilized hole-selective layer for high-performance inverted p-i-n perovskite solar cells. Science *382*, 284–289. https://doi.org/10.1126/science.ade9637.

20. Liu, M., Bi, L., Jiang, W., Zeng, Z., Tsang, S.-W., Lin, F.R., and Jen, A.K.-Y. (2023). Compact Hole-Selective Self-Assembled Monolayers Enabled by Disassembling Micelles in Solution for Efficient Perovskite Solar Cells. Adv. Mater. *35*, e2304415. https://doi.org/10.1002/adma.202304415.

21. Cui, P., Wei, D., Ji, J., Huang, H., Jia, E., Dou, S., Wang, T., Wang, W., and Li, M. (2019). Planar p–n homojunction perovskite solar cells with efficiency exceeding 21.3%. Nat. Energy *4*, 150–159. https://doi.org/10.1038/s41560-018-0324-8.

22. Wang, Q., Shao, Y., Xie, H., Lyu, L., Liu, X., Gao, Y., and Huang, J. (2014). Qualifying composition dependent p and n self-doping in CH3NH3PbI3. Appl. Phys. Lett. *105*, 163508. https://doi.org/10.1063/1.4899051.

23. Qu, G., Dong, L., Qiao, Y., Khan, D., Chen, Q., Xie, P., Yu, X., Liu, X., Wang, Y., Chen, J., et al. (2022). Dopant-Free Phthalocyanine Hole Conductor with Thermal-Induced Holistic Passivation for Stable Perovskite Solar Cells with 23% Efficiency. Adv. Funct. Materials *32*, 2206585. https://doi.org/10.1002/adfm.202206585.

24. Gong, S., Huang, Y., Yu, X., Hu, Q., Liu, J., Meng, J., Wen, Y., and Chen, X. (2023). Ultrafast dynamics in perovskite-based optoelectronic devices. Cell Reports Physical Science *4*, 101580. https://doi.org/10.1016/j.xcrp.2023.101580.

25. Wang, J., Zhang, J., Zhou, Y., Liu, H., Xue, Q., Li, X., Chueh, C.-C., Yip, H.-L., Zhu, Z., and Jen, A.K.Y. (2020). Highly efficient all-inorganic perovskite solar cells with suppressed non-radiative recombination by a Lewis base. Nat. Commun. *11*, 177. https://doi.org/10.1038/s41467-019-13909-5.

26. Green, M.A. (1982). Accuracy of analytical expressions for solar cell fill factors. Sol. Cells *7*, 337–340. https://doi.org/10.1016/0379-6787(82)90057-6.

27. Li, X., Wu, X., Li, B., Cen, Z., Shang, Y., Lian, W., Cao, R., Jia, L., Li, Z., Gao, D., et al. (2022). Modulating the deep-level defects and charge extraction for efficient perovskite solar cells with high fill factor over 86%. Energy Environ. Sci. *15*, 4813–4822. https://doi.org/10.1039/D2EE02543D.

28. Cao, Q., Li, Y., Zhang, H., Yang, J., Han, J., Xu, T., Wang, S., Wang, Z., Gao, B., Zhao, J., et al. (2021). Efficient and stable inverted perovskite solar cells with very high fill factors via incorporation of star-shaped polymer. Sci. Adv. *7*, eabg0633. https://doi.org/10.1126/sciadv.abg0633.

29. Rai, M., Wong, L.H., and Etgar, L. (2020). Effect of Perovskite Thickness on Electroluminescence and Solar Cell Conversion Efficiency. J. Phys. Chem. Lett. *11*, 8189–8194. https://doi.org/10.1021/acs.jpclett.0c02363.

30. Khenkin, M.V., Katz, E.A., Abate, A., Bardizza, G., Berry, J.J., Brabec, C., Brunetti, F., Bulović, V., Burlingame, Q., Di Carlo, A., et al. (2020). Consensus statement for stability assessment and reporting for perovskite photovoltaics based on ISOS procedures. Nat. Energy *5*, 35–49. https://doi.org/10.1038/s41560-019-0529-5.